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SEMICONDUCTOR PACKAGE INCLUDING PACKAGE SUBSTRATE WITH DUMMY VIA AND METHOD OF FORMING THE SAME

Abstract

A semiconductor package may include a package substrate including a dummy via on a first side of the package substrate, an interposer module on a second side of the package substrate opposite the first side of the package substrate, and a stiffener ring on the second side of the package substrate and including an edge that is substantially aligned with the dummy via.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. Non-provisional patent application Ser. No. 17/862,040 entitled “Semiconductor Package Including Package Substrate with Dummy Via and Method of Forming the Same” filed Jul. 11, 2022, the entire contents of which are incorporated herein by reference for all purposes.

BACKGROUND

[0002] Interfaces between a fan-out wafer level package (FOWLP) and an underfill material portion are subjected to mechanical stress during subsequent handling of an assembly of the FOWLP, the underfill material portion, and a packaging substrate, such as the mechanical stress associated with attaching the packaging substrate to a printed circuit board (PCB). In addition, interfaces between a FOWLP and an underfill material portion are subjected to mechanical stress during use within a computing device, such as when a mobile device is accidentally dropped to cause a mechanical shock during usage. Cracks may be formed in the underfill material, and may induce additional cracks in a semiconductor die, solder material portions, interposers, and/or various dielectric layers within a semiconductor die or within a packaging substrate. A semiconductor package may typically include one or more semiconductor devices (e.g., semiconductor die, system on chip (SOC) device, high bandwidth memory (HBM) device) that are mounted on a package substrate. Thus, suppression of the formation of cracks in the underfill material and other package components is desired. A stiffener ring or stiffener lid may be formed on the package substrate in order to provide rigidity to the package substrate and mitigate against cracks at various levels in the semiconductor package.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1A is a vertical cross-sectional view of a semiconductor package according to one or more embodiments.

[0005] FIG. 1B is a plan view of the board-side surface of package substrate lower dielectric layer, according to one or more embodiments.

[0006] FIG. 2A is a vertical cross-sectional view of an intermediate structure including a core material layer according to one or more embodiments.

[0007] FIG. 2B is a vertical cross-sectional view of an intermediate structure including openings in the core material layer according to one or more embodiments.

[0008] FIG. 2C is a vertical cross-sectional view of an intermediate structure including the through via in the core according to one or more embodiments.

[0009] FIG. 2D is a vertical cross-sectional view of an intermediate structure including the package

substrate upper dielectric layer according to one or more embodiments.

[0010] FIG. 2E is a vertical cross-sectional view of an intermediate structure including the package substrate lower dielectric layer according to one or more embodiments.

[0011] FIG. 2F is a vertical cross-sectional view of an intermediate structure including openings according to one or more embodiments.

[0012] FIG. 2G is a vertical cross-sectional view of an intermediate structure including dummy vias according to one or more embodiments.

[0013] FIG. 2H illustrates a vertical cross-sectional view of an intermediate structure in which the interposer module may be mounted on the package substrate (e.g., via a flip chip bonding (FCB) process) according to one or more embodiments.

[0014] FIG. 2I illustrates a vertical cross-sectional view of an intermediate structure in which the package underfill layer may be formed on the package substrate according to one or more embodiments.

[0015] FIG. 2J illustrates a vertical cross-sectional view of an intermediate structure in which the adhesive may be applied to the package substrate according to one or more embodiments.

[0016] FIG. 2K illustrates a vertical cross-sectional view of an intermediate structure in which the stiffener ring may be attached to (e.g., mounted on) the package substrate according to one or more embodiments.

[0017] FIG. 2L illustrates a vertical cross-sectional view of an intermediate structure in which a plurality of solder balls may be formed on the package substrate according to one or more embodiments.

[0018] FIG. 3 illustrates a vertical cross-sectional view of a first alternative design of the semiconductor package, according to one or more embodiments.

[0019] FIG. 4A illustrates a vertical cross-sectional view of an intermediate structure in which an opening has been formed according to one or more embodiments.

[0020] FIG. 4B illustrates a vertical cross-sectional view of an intermediate structure in which dummy via layer has been formed according to one or more embodiments.

[0021] FIG. 4C illustrates a vertical cross-sectional view of an intermediate structure in which an opening has been formed according to one or more embodiments.

[0022] FIG. 4D illustrates a vertical cross-sectional view of an intermediate structure in which dummy via layer has been formed according to one or more embodiments.

[0023] FIG. 5 illustrates a vertical cross-sectional view of a second alternative design of the semiconductor package, according to one or more embodiments.

[0024] FIG. 6A illustrates a vertical cross-sectional view of a third alternative design of the semiconductor package, according to one or more embodiments.

[0025] FIG. 6B is a plan view of the board-side surface of package substrate lower dielectric layer for the third alternative design of the semiconductor package, according to one or more embodiments.

[0026] FIG. 7 is a plan view of the board-side surface of package substrate lower dielectric layer for a fourth alternative design of the semiconductor package, according to one or more embodiments.

[0027] FIG. 8A illustrates the dummy vias having a line shaped cross-section (e.g., looking down onto the board-side surface of the package substrate lower dielectric layer), according to one or more embodiments.

[0028] FIG. 8B illustrates the dummy vias having a rectangular shaped cross-section (e.g., square shaped), according to one or more embodiments.

[0029] FIG. 8C illustrates the dummy vias having a hexagonal shaped cross-section, according to one or more embodiments.

[0030] FIG. 9 is a flow chart illustrating a method of making a semiconductor package according to one or more embodiments.

DETAILED DESCRIPTION

[0031] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0032] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Unless explicitly stated otherwise, each element having the same reference numeral is presumed to have the same material composition and to have a thickness within a same thickness range.

[0033] In a related semiconductor package, after a thermal cycle reliability test, a crack in the package substrate may be detected. In particular, the crack may be detected in the package substrate on a ball grid array (BGA) side of a dielectric layer (e.g., Ajinomoto Build-up Film (ABF) layer) around an edge of a stiffener ring. The crack may penetrate into the dielectric layer and stop at the core of the package substrate, or at a via structure in the core. The crack may be particularly common near a metal trace (e.g., copper trace) in the package substrate.

[0034] A possible cause of the crack may include a strong stiffener ring design. The stiffener ring may be intended to constrain warping of the package. However, the stiffener ring may induce enhanced tensile stress on the BGA side of the dielectric layer of the package substrate.

[0035] One or more embodiments of the present disclosure may include a semiconductor package (e.g., organic/silicon interposer package, semiconductor device) with an enhanced substrate design. In particular, the package substrate may include one or more dummy vias that may mitigate development of a crack in the package substrate. The dummy vias may be arranged, for example, in an array (e.g., having rows and columns). More particularly, the one or more dummy vias may be located on the BGA side of the dielectric layer on substrate, and may enhance a structure of the semiconductor package, and thereby, inhibit (e.g., prevent) a crack issue (e.g., metal trace crack issue) in the package substrate. The various embodiment substrate designs may improve reliability of the semiconductor package.

[0036] The semiconductor package may include, for example, a stiffener ring or a stiffener lid on a surface of the package substrate. The one or more dummy vias (e.g., dummy via array) may be located close to an edge of the stiffener ring. In particular, a centerline of the one or more dummy vias may be substantially aligned with the edge of the stiffener ring. The stiffener lid may include a ring and a cap integrated together. The ring may surround the semiconductor device and the cap may cover the ring and the semiconductor device.

[0037] In one or more embodiments, the centerline of the one or more dummy vias may be located by a distance D from an edge of the stiffener foot that is greater than 0.1 mm and less than 2 mm ($0.1 < D < 2$ mm). The one or more dummy vias may have either a stacked (e.g., multi-stacked such as 2-stacking, 3-stacking, etc.) or unstacked structure. A total height A of the one or more dummy vias may be less than a thickness B of the dielectric layer on the BGA side of the package substrate

($A < B$). An area ratio of the one or more dummy vias may be greater than 0.1 and less than 0.9. That is, where the inner edge of the stiffener ring has a length L and a width W , the following may be satisfied: $0.1 < \Sigma C / [2D * (L + W)] < 0.9$, where C is the area of each dummy via in a via array; D is the distance from the centerline of the one or more dummy vias (which may be coincident with an edge of the stiffener foot) to the edge of the dummy via or edge of the dummy via array; L is the length of the inner edge of the stiffener ring **150**; and W is the width of the inner edge of the stiffener ring **150**. In some embodiments, D may be the radius of the dummy via. In other embodiments, D may be the distance from the centerline of the dummy via array to the edge of the dummy via array.

[0038] FIG. **1A** is a vertical cross-sectional view of a semiconductor package **100** according to one or more embodiments. Generally, the semiconductor package **100** may include a package substrate **110**, an interposer module **120** on the package substrate **110**, and a stiffener ring **150** adhered and/or affixed to the package substrate **110** adjacent to the interposer module **120**. The stiffener ring **150** may include an inner edge **150a** and an outer edge **150b**.

[0039] The package substrate **110** may include, for example, a core **112**, a package substrate upper dielectric layer **114** formed on the core **112** (e.g., a first side or chip-side of the package substrate **110**), and a package substrate lower dielectric layer **116** formed on the core **112** (e.g., a second side or board-side of the package substrate **110**). In particular, the package substrate **110** may include a build-up film substrate such as an Ajinomoto build-up film (ABF) substrate. That is, in at least one embodiment, each of the package substrate upper dielectric layer **114** and the package substrate lower dielectric layer **116** may be described as an ABF layer.

[0040] The core **112** may help to provide rigidity to the package substrate **110**. The core **112** may include, for example, an epoxy resin such as a bismaleimide triazine epoxy (BT epoxy) and/or a woven glass laminate. The core **112** may alternatively or in addition include an organic material such as a polymer material. In particular, the core **112** may include a dielectric polymer material such as polyimide (PI), benzocyclobutene (BCB), or polybenzobisoxazole (PBO). Other suitable dielectric materials are within the contemplated scope of disclosure.

[0041] The core **112** may include one or more through vias **112a**. The through vias **112a** may extend from a lower surface of the core **112** to an upper surface of the core **112**. The through vias **112a** may allow an electrical connection between the package substrate upper dielectric layer **114** and the package substrate lower dielectric layer **116**. The through vias **112a** may include, for example, one or more layers and may include metals, metal alloys, and/or other metal-containing compounds (e.g., Cu, Al, Mo, Co, Ru, W, TiN, TaN, WN, etc.). Other suitable metal materials are within the contemplated scope of disclosure.

[0042] The package substrate upper dielectric layer **114** may be formed on an upper surface of the core **112**. The package substrate upper dielectric layer **114** may include a plurality of layers and, in particular, may include a build-up film (e.g., ABF). The package substrate upper dielectric layer **114** may also include an organic material such as a polymer material. In particular, the package substrate upper dielectric layer **114** may include a dielectric polymer material such as polyimide (PI), benzocyclo-butene (BCB), or polybenzobisoxazole (PBO). Other suitable dielectric materials are within the contemplated scope of disclosure.

[0043] The package substrate upper dielectric layer **114** may include one or more package substrate upper bonding pads **114a** on a chip-side surface of the package substrate upper dielectric layer **114**. In particular, the package substrate upper bonding pads **114a** may be exposed on the chip-side surface of the package substrate upper dielectric layer **114**. The package substrate upper dielectric layer **114** may also include one or more metal interconnect structures **114b**. The metal interconnect structures **114b** may be connected to the package substrate upper bonding pads **114a** and the through vias **112a** in the core **112**. The metal interconnect structures **114b** may include metal layers (e.g., copper traces) and metal vias connecting the metal layers. The package substrate upper bonding pads **114a** and the metal interconnect structures **114b** may include, for example, one or

more layers and may include metals, metal alloys, and/or other metal-containing compounds (e.g., Cu, Al, Mo, Co, Ru, W, TiN, TaN, WN, etc.). Other suitable metal materials are within the contemplated scope of disclosure.

[0044] An upper passivation layer **110a** may be formed on the chip-side surface of the package substrate upper dielectric layer **114**. The upper passivation layer **110a** may partially cover the package substrate upper bonding pads **114a**. The upper passivation layer **114a** may include silicon oxide, silicon nitride, low-k dielectric materials such as carbon-doped oxides, extremely low-k dielectric materials such as porous carbon doped silicon dioxide, a combination thereof or other suitable material.

[0045] The package substrate lower dielectric layer **116** may be formed on an lower surface of the core **112**. The package substrate lower dielectric layer **116** may also include a plurality of layers and, in particular, may include a build-up film (e.g., ABF). The package substrate lower dielectric layer **116** may also include an organic material such as a polymer material. In particular, the package substrate lower dielectric layer **116** may include a dielectric polymer material such as polyimide (PI), benzocyclobutene (BCB), or polybenzobisoxazole (PBO). Other suitable dielectric materials are within the contemplated scope of disclosure.

[0046] The package substrate lower dielectric layer **116** may include one or more package substrate lower bonding pads **116a** on a board-side surface of the package substrate lower dielectric layer **116**. In particular, the package substrate lower bonding pads **116a** may be exposed on the chip-side surface of the package substrate lower dielectric layer **116**. The package substrate lower dielectric layer **116** may also include one or more metal interconnect structures **116b**. The metal interconnect structures **116b** may be connected to the package substrate lower bonding pads **116a** and the through vias **112a** in the core **112**. The metal interconnect structures **116b** may include metal layers (e.g., copper traces) and metal vias connecting the metal layers. The package substrate upper bonding pads **116a** and the metal interconnect structures **116b** may include, for example, one or more layers and may include metals, metal alloys, and/or other metal-containing compounds (e.g., Cu, Al, Mo, Co, Ru, W, TiN, TaN, WN, etc.). Other suitable metal materials are within the contemplated scope of disclosure.

[0047] A lower passivation layer **110b** may be formed on the board-side surface of the package substrate lower dielectric layer **116**. The lower passivation layer **110b** may partially cover the package substrate lower bonding pads **116a**. The lower passivation layer **110a** may include silicon oxide, silicon nitride, low-k dielectric materials such as carbon-doped oxides, extremely low-k dielectric materials such as porous carbon doped silicon dioxide, a combination thereof or other suitable material.

[0048] The package substrate lower dielectric layer **116** may also include one or more dummy vias **116c**. The dummy vias **116c** may be formed in the board-side surface of the package substrate lower dielectric layer **116** and may or may not be exposed at the board-side surface of the package substrate lower dielectric layer **116**. A height of the dummy vias **116c** in the z-direction may be less than a thickness of the package substrate lower dielectric layer **116**. That is, a thickness of the package substrate lower dielectric layer **116** may be greater than a thickness of the dummy vias **116**.

[0049] The dummy vias **116c** may be aligned with an edge of the stiffener ring **150**. In particular, as illustrated in FIG. 1A, a centerline in the x-direction (as indicated by the dashed line in FIG. 1A) of the dummy via **116c** may be substantially aligned with the inner edge **150a** of the stiffener ring **150**. Although it is not illustrated in FIG. 1A, at least some portion of the dummy vias **116c** may be located underneath the interposer module **120** in the z-direction.

[0050] A lowermost surface of the dummy vias **116c** may be substantially coplanar with the lowermost surface of the package substrate lower bonding pads **116a** and with the board-side surface of the package substrate lower dielectric layer **116**. The dummy vias **116c** may be exposed at the board-side surface of the package substrate lower dielectric layer **116**, or may be covered by

another layer of the package substrate **110** such as the lower passivation layer **110b**.

[0051] A ball-grid array (BGA) including a plurality of solder balls **110c** may be formed on the board-side surface of the package substrate lower dielectric layer **116**. The solder balls **110c** may allow the semiconductor package **100** to be securely mounted on a substrate such as a printed circuit board (PCB) and electrically coupled to the PCB substrate. The solder balls **110c** may contact the package substrate lower bonding pads **116a**, respectively. The solder balls **110c** may therefore be electrically connected to the package substrate upper bonding pads **114a** by way of metal interconnect structures **116b**, the through vias **112a** and the metal interconnect structures **114b**.

[0052] The interposer module **120** may be mounted by C4 bumps **121** on the package substrate upper bonding pads **114a** in the package substrate **110**. The interposer module **120** may include an interposer **122** which may include an organic material (e.g., dielectric polymer) or inorganic material (e.g., silicon). In particular, as illustrated in FIG. 1A, the interposer **122** may include a plurality of layers (e.g., dielectric polymer layers, silicon layers, etc.). The interposer **122** may include metal interconnects **122a** connected to the C4 bumps **121**.

[0053] The interposer module **120** may also include one or more semiconductor dies (e.g., semiconductor chips) mounted on the interposer **122**. In particular, as illustrated in FIG. 1A, a first semiconductor die **143** and second semiconductor die **144** may be mounted on the interposer **122**. The first semiconductor die **143** and second semiconductor die **144** may be mounted on the interposer **122**, for example, by micro-bumps **128** that may be electrically connected to the metal interconnects **122a** in the interposer **122**. A second package underfill layer **129** may be formed under and around the interposer module **120** and the C4 bumps **121** so as to fix the interposer module **120** to the package substrate **110**. The package underfill layer **129** may be formed of an epoxy-based polymeric material.

[0054] Each of the first semiconductor die **143** and second semiconductor die **144** may include, for example, a semiconductor die, a system on chip (SOC) die, a system on integrated chips (SoIC) die, a high-bandwidth memory (HBM) die and a dynamic random access memory (DRAM) die. In particular, the interposer module **120** may include a high-performance computing (HPC) application and may include, for example, an integrated graphics processing unit (GPU), application specific integrated circuit (ASIC), field-programmable gate array (FPGA), and HBM by chip on wafer on substrate (CoWoS) technology or integrated fan-out on substrate (INFO-oS) technology.

[0055] An interposer underfill layer **126** may be formed around the micro-bumps **128** and between the first semiconductor die **143** and the interposer **122** and between the second semiconductor die **144** and the interposer **122**. The interposer underfill layer **126** may be formed as two separate portions under the first semiconductor die **143** and second semiconductor die **144** respectively. Alternatively, the interposer underfill layer **126** may be formed continuously under both of the first semiconductor die **143** and second semiconductor die **144**. The interposer underfill layer **126** may also be formed between first semiconductor die **143** and the second semiconductor die **144**. The interposer underfill layer **126** may also be formed of an epoxy-based polymeric material.

[0056] A molding material layer **127** may be formed over the first semiconductor die **143**, the second semiconductor die **144**, the interposer underfill layer **126** and the interposer **122**. The molding material layer **127** may be formed of an epoxy molding compound (EMC).

[0057] The stiffener ring **150** may be mounted on the package substrate **110** around the interposer module **120**. The stiffener ring **150** may be securely fixed to the package substrate **110** by an adhesive **160** (e.g., a silicone adhesive or an epoxy adhesive). The stiffener ring **150** may be formed of a metal such as copper with a nickel coating, or an aluminum alloy. The stiffener ring **150** may provide rigidity to the package substrate **110**.

[0058] FIG. 1B is a plan view of the board-side surface of package substrate lower dielectric layer **116**, according to one or more embodiments. A location of the stiffener ring **150** and the interposer

module **120** on the opposing surface (e.g., chip-side surface) of the package substrate **110** is indicated by shading. The outer periphery of the interposer module **120** and the inner edge **150a** and outer edge **150b** of the stiffener ring **150** are also indicated by a dashed line in FIG. **1B**.

[0059] As illustrated in FIG. **1B**, the stiffener ring **150** may be located around an entire periphery of the interposer module **120**. The solder balls **110c** of the BGA may be located under the stiffener ring **150** and under the interposer module **120**.

[0060] The dummy vias **116c** may be located adjacent to solder balls **110c** of the BGA. In particular, the dummy vias **116c** may be formed between an inner set of solder balls **110c** under the interposer module **120**, and an outer set of solder balls **110c** under the stiffener ring **150**.

[0061] The dummy vias **116c** may be formed around an entire periphery of the interposer module **120** in the plan view. In particular, the dummy vias **116c** may be formed under the inner edge **150a** of the stiffener ring **150** over the entire length of the inner edge **150a**. The inner edge **150a** may have a length L in the x-direction and a width W in the y-direction. Therefore, a total length of the inner edge **150a** may be given by $2(L+W)$.

[0062] The dummy vias **116c** may be substantially aligned under the inner edge **150a** of the stiffener ring **150**. Thus, in a plan view the dummy vias **116c** may be arranged so as to have a ring shape substantially the same as a ring shape of the stiffener ring **150**.

[0063] As illustrated in FIG. **1B**, the centerline of the dummy vias **116c** may be coincident with inner edge **150a** of the stiffener ring **150**. The dummy vias **116c** may extend away from the centerline by a distance D . Thus, a width (e.g., diameter) of the dummy vias **116c** in the x-direction and y-direction may be $2D$. In at least one embodiment, the distance D from the centerline of the dummy vias **116c** to an edge of the dummy vias **116c** may be greater than 0.1 mm and less than 2 mm .

[0064] The dummy vias **116c** may be located in a region R having a width that is substantially the same as the width ($2D$) of the dummy vias **116c**.

[0065] It may be preferable to set a density of the dummy vias **116c** in the region R . This may be done by setting an area ratio R_a for the dummy vias **116c**. The area ratio R_a may be defined as the ratio of the total area ADV of the dummy vias **116c** to the area AR of a region R in which the dummy vias **116c** are located (i.e., $R_a=ADV/AR$). The area ratio R_a may be, for example, greater than 0.1 and less than 0.9 .

[0066] For example, in FIG. **1B**, there may be a total of 46 dummy vias **116c**. Each dummy via **116c** may have a circular horizontal cross sectional profile with a radius r (note that in this particular example, the radius r may be equal to the distance D). Thus, the total area ADV of the dummy vias **116c** would be given by $46\pi r^2$. The area AR of region R may be given by $2D \times 2(L+W)$. Thus, the area ratio R_a would be given by $46\pi r^2/[4D(L+W)]$.

[0067] FIG. **2A** is a vertical cross-sectional view of an intermediate structure including a core material layer **112L** according to one or more embodiments. The core **112** may be formed by first preparing a core material layer **112L**. The core material layer **112L** may include a layer or plurality of layers of epoxy resin and/or a woven glass laminate. Lower via pads **112a-1** may then be formed on the core material layer **112L**. To form the lower via pads **112a-1**, a layer of metal material (e.g., copper, aluminum or other suitable conductive materials) may be deposited (e.g., by CVD, PVD or other suitable deposition technique) on the core material layer **112L**. The metal layer may be etched (e.g., by wet etching, dry etching, etc.) so as to form the metal bonding pads **112a-1**.

[0068] FIG. **2B** is a vertical cross-sectional view of an intermediate structure including openings O in the core material layer **112L** according to one or more embodiments. The openings O may be formed in the core material layer **112L** at locations corresponding to the lower via pads **112a-1**. The openings O may be formed by using a photolithographic process. The photolithographic process may include forming a patterned photoresist mask (not shown) on the core material layer **112L**, and etching (e.g., wet etching, dry etching, etc.) the exposed upper surface of the core material layer **112L** through openings in the photoresist mask. The photoresist mask may be subsequently

removed by ashing, dissolving the photoresist mask or by consuming the photoresist mask during the etch process.

[0069] FIG. 2C is a vertical cross-sectional view of an intermediate structure including the through via **112a** in the core **112** according to one or more embodiments. Via columns **112a-2** and upper via pads **112a-3** may then be formed in the same step or consecutive steps, to complete the formation of the through vias **112a** and the core **112**. To form the via columns **112a-2** and upper via pads **112a-3** in one step, for example, a metal layer (e.g., copper, aluminum or other suitable conductive materials) may be deposited in the openings **O** and on an upper surface of the core material layer **112L** by CVD, PVD or other suitable deposition technique. The metal layer may be etched (e.g., by wet etching, dry etching, etc.) so as to form the via columns **112a-2** and upper via pads **112a-3**.

[0070] FIG. 2D is a vertical cross-sectional view of an intermediate structure including the package substrate upper dielectric layer **114** according to one or more embodiments. The package substrate upper dielectric layer **114** may include a plurality of dielectric layers **114** that may be formed, for example, by chemical vapor deposition (CVD), physical vapor deposition (PVD), spin coating, lamination or other suitable deposition technique.

[0071] The metal layers (e.g., traces) and metal vias of the metal interconnect structures **114b** may be formed layer-by-layer at the time of forming the plurality of dielectric layers of the package substrate upper dielectric layer **114**. The metal interconnect structures **114b** may be formed so as to contact the through vias **112a** in the core **112**. In particular, the metal layers (e.g., traces) and metal vias of the metal interconnect structures **114b** may be formed by a semi-additive process (SAP).

[0072] The package substrate upper bonding pads **114a** may be formed in an uppermost dielectric layer of the package substrate upper dielectric layer **114**. The package substrate upper bonding pads **114a** may be formed so as to contact the metal interconnect structures **114b**. The package substrate upper bonding pads **114a** may be formed by depositing a metal layer (e.g., copper, aluminum or other suitable conductive materials) on a dielectric layer of the package substrate upper dielectric layer **114**. The metal layer may then be patterned by etching (e.g., by wet etching, dry etching, etc.) so as to form the package substrate upper bonding pads **114a**.

[0073] The upper passivation layer **110a** may then be formed on the surface of the uppermost dielectric layer of the package substrate upper dielectric layer **114**. The upper passivation layer **110a** may be formed so as to partially cover the package substrate upper bonding pads **114a**. The upper passivation layer **110a** may be formed, for example, by chemical vapor deposition (CVD), physical vapor deposition (PVD), spin coating, lamination or other suitable deposition technique.

[0074] FIG. 2E is a vertical cross-sectional view of an intermediate structure including the package substrate lower dielectric layer **116** according to one or more embodiments. The package substrate lower dielectric layer **116** may be formed by a process that is similar to the process of forming package substrate upper dielectric layer **114**. The package substrate lower dielectric layer **116** may include a plurality of dielectric layers that may be formed, for example, by chemical vapor deposition (CVD), physical vapor deposition (PVD), spin coating, lamination or other suitable deposition technique.

[0075] The metal layers (e.g., traces) and metal vias of the metal interconnect structures **116b** may be formed layer-by-layer at the time of forming the plurality of dielectric layers of the package substrate lower dielectric layer **116**. The metal interconnect structures **116b** may also be formed so as to contact the through vias **112a** in the core **112**. In particular, the metal layers (e.g., traces) and metal vias of the metal interconnect structures **116b** may be formed by a semi-additive process (SAP).

[0076] The package substrate lower bonding pads **116a** may be formed in an uppermost dielectric layer of the package substrate upper dielectric layer **116**. The package substrate lower bonding pads **116a** may be formed at locations corresponding to a location of the solder balls **110c** of the BGA. The package substrate lower bonding pads **116a** may also be formed so as to contact the metal interconnect structures **116b**. The package substrate lower bonding pads **116a** may be formed by

depositing a metal layer (e.g., copper, aluminum or other suitable conductive materials) on a dielectric layer of the package substrate lower dielectric layer **116**. The metal layer may then be patterned by etching (e.g., by wet etching, dry etching, etc.) so as to form the package substrate lower bonding pads **116a**.

[0077] FIG. 2F is a vertical cross-sectional view of an intermediate structure including openings **116c-O** according to one or more embodiments. The openings **116c-O** may be formed in the package substrate lower dielectric layer **116** in locations corresponding to the dummy vias **116c**. In particular, the openings **116c-O** may be formed in the region R (e.g., see FIG. 1B) around an entire periphery of the interposer module **120**, between the lower bonding pads **116a**, between the metal interconnect structures **116b**, and so as to have a centerline (e.g., in the x-direction) corresponding to an inner edge **150a** of the stiffener ring **150**. The openings **116c-O** may also be formed so as to have a size (e.g., depth, diameter, length, width, etc.) and shape (e.g., circular cylinder, conical cylinder) corresponding to the size and shape of the dummy vias **116c**. In particular, the distance D from the centerline of the openings **116c-O** to an edge of the openings **116c-O** may be greater than 0.1 mm and less than 2 mm. In addition, a depth of the openings **116c-O** may be no greater than a total thickness of the package substrate lower dielectric layer **116**. The openings **116c-O** may be formed by using a photolithographic process. The photolithographic process may include forming a patterned photoresist mask (not shown) on an upper surface of the package substrate lower dielectric layer **116**, and etching (e.g., wet etching, dry etching, etc.) the exposed upper surface through openings in the photoresist mask. The photoresist mask may be subsequently removed by ashing, dissolving the photoresist mask or by consuming the photoresist mask during the etch process.

[0078] FIG. 2G is a vertical cross-sectional view of an intermediate structure including dummy vias **116c** according to one or more embodiments. The dummy vias **116c** may be formed, for example, by depositing a metal layer (e.g., copper, aluminum or other suitable conductive materials) in the openings **116c-O** and on the upper surface of the package substrate lower dielectric layer **114** by CVD, PVD or other suitable deposition technique. The metal layer may then be etched (e.g., by wet etching, dry etching, etc.) so as to form the dummy vias **116**.

[0079] The lower passivation layer **110b** may then be formed on the upper surface of the package substrate lower dielectric layer **116**. The lower passivation layer **110b** may be formed so as to partially cover the package substrate lower bonding pads **116a**. The lower passivation layer **110b** may or may not cover the dummy vias **116c**. The lower passivation layer **110b** may be formed, for example, by chemical vapor deposition (CVD), physical vapor deposition (PVD), spin coating, lamination or other suitable deposition technique.

[0080] FIG. 2H illustrates a vertical cross-sectional view of an intermediate structure in which the interposer module **120** may be mounted on the package substrate **110** (e.g., via a flip chip bonding (FCB) process) according to one or more embodiments. As illustrated in FIG. 2A, a solder portion of the C4 bumps **121** of the interposer module **120** may be positioned on the package substrate upper bonding pads **114a** on the package substrate **110**. The intermediate structure may then be heated in order to bond the solder portion of the C4 bumps **121** to a part of the package substrate upper bonding pads **114a** that is exposed through the upper passivation layer **110a**.

[0081] FIG. 2I illustrates a vertical cross-sectional view of an intermediate structure in which the package underfill layer **129** may be formed on the package substrate **110** according to one or more embodiments. The package underfill layer **129** may be formed of an epoxy-based polymeric material. As illustrated in FIG. 2I, the package underfill layer **129** may be formed under and around the interposer module **120** and the C4 bumps **121** so as to fix the interposer module **120** to the package substrate **110**. The package underfill layer **129** may then be cured, for example, in a box oven for about 90 minutes at about 150° C. to provide the package underfill layer **129** with a sufficient stiffness and mechanical strength.

[0082] FIG. 2J illustrates a vertical cross-sectional view of an intermediate structure in which the

adhesive **160** may be applied to the package substrate **110** according to one or more embodiments. The adhesive **160** may include, for example, a silicone adhesive or an epoxy adhesive. The adhesive **160** may be located on the package substrate **110** at a position corresponding to a placement of the stiffener ring **150**. In particular, the adhesive **160** may be formed as a continuous bead around an entire periphery of the interposer module **120**. The adhesive **160** may be dispensed on a surface of the package substrate **110** in a quantity sufficient to securely bond the stiffener ring **150** to the package substrate **110**.

[0083] FIG. 2K illustrates a vertical cross-sectional view of an intermediate structure in which the stiffener ring **150** may be attached to (e.g., mounted on) the package substrate **110** according to one or more embodiments. The stiffener ring **150** may be composed of metal material (e.g., aluminum) and may be formed, for example, by milling using a computer numerical control (CNC) milling machine.

[0084] The package substrate **110** with the interposer module **120** may be placed on a surface and the stiffener ring **150** lowered down onto the package substrate **110** around the interposer module **120**. The stiffener ring **150** may then be aligned with the adhesive **160** formed on the package substrate **110**. The stiffener ring **150** may then be pressed downward by applying a pressing force down onto the stiffener ring **150** so that the stiffener ring **150** may be fixed to the package substrate **110** through the adhesive **160**.

[0085] Alternatively, the stiffener ring **150** may be placed on a surface (e.g., a flat surface), and the package substrate **110** inverted and lowered onto the stiffener ring **150**. That is, the interposer module **120** is inserted into the stiffener ring **150**. The package substrate **110** and interposer module **120** may then be pressed by applying a pressing force down into stiffener ring **150** so that the stiffener ring **150** is fixed to the package substrate **110** through the adhesive **160**.

[0086] The stiffener ring **150** may be clamped to the package substrate **110** for a period to allow the adhesive **160** to cure and form a secure bond between the package substrate **110** and the stiffener ring **150**. The clamping of the stiffener ring **150** to the package substrate **110** may be performed, for example, by using a heat clamp module. The heat clamp module may apply a uniform force across the upper surface of the stiffener ring **150**.

[0087] FIG. 2L illustrates a vertical cross-sectional view of an intermediate structure in which a plurality of solder balls **110c** may be formed on the package substrate **110** according to one or more embodiments. The plurality of solder balls **110c** may be formed on the lower bonding pads **116a**, for example, by an electroplating process. The plurality of solder balls **110c** may contact the lower bonding pads **116a** through openings in the lower passivation layer **110b**. The plurality of solder balls **110c** may constitute a ball-grid array (BGA) that may allow the semiconductor package **100** to be securely mounted (e.g., by surface mount technology (SMT)) on a substrate such as a printed circuit board and electrically coupled to the substrate.

[0088] The solder balls **110c** may be formed, for example, so as to be located under the stiffener ring **150** and under the interposer module **120**, and so as to be located adjacent to the dummy vias **116c**. In particular, the solder balls **110c** located under the interposer module **120** and solder balls **110c** under the stiffener ring **150** may be located on opposing sides of the dummy vias **116c**.

[0089] FIG. 3 illustrates a vertical cross-sectional view of a first alternative design of the semiconductor package **100**, according to one or more embodiments. As illustrated in FIG. 3, the first alternative design may include stacked dummy vias **116d**. Each stacked dummy via **116d** may include a dummy via layer **116c2** and a dummy via layer **116c3** stacked on the dummy via layer **116c2**. It should be noted that although two dummy via layers are illustrated in FIG. 3, the stacked dummy vias **116d** in the first alternative design may include any number of stacked dummy via layers.

[0090] The stacked dummy vias **116d** may be substantially similar to the dummy vias **116c** described above. In particular, the stacked dummy vias **116d** may have a structure (e.g., size and shape) and function that are substantially similar to the structure and function of the dummy vias

116c described above (e.g., see FIG. 1A). The stacked dummy vias **116d** may also have an arrangement (e.g., layout) that is substantially similar to the arrangement of the dummy vias **116c** described above (e.g., see FIG. 1B).

[0091] An advantage of the first alternative design in FIG. 3 is that the stacked dummy vias **116d** may be formed during a formation of the package substrate lower dielectric layer **116**. This may result in fewer overall processing steps. The stacked dummy vias **116d** may be formed in the same dielectric layers and using the same materials that are used in forming the metal interconnect structures **116b**. A thickness of each of the dummy via layers (e.g., dummy via layer **116c2** and a dummy via layer **116c3**) may be substantially the same as a thickness of the dielectric layer in which the dummy via layer (e.g., dummy via layer **116c2**, dummy via layer **116c3**) is formed.

[0092] FIGS. 4A-4D illustrate a method of forming the stacked dummy via **116d** in the first alternative design, according to one or more embodiments. In particular, FIGS. 4A-4D illustrate a detailed view of the Region A in FIG. 3.

[0093] FIG. 4A illustrates a vertical cross-sectional view of an intermediate structure in which an opening **116c-02** has been formed according to one or more embodiments. As illustrated in FIG. 4A, a first dielectric layer **116L-1** of the package substrate lower dielectric layer **116** may be formed on the core **112** by any suitable deposition method (e.g., CVD, PVD, etc.). A metal material (e.g., metal trace) may be formed on the first dielectric layer **116L-1** by any suitable deposition method (e.g., CVD, PVD, etc.) and patterned by a photolithographic process to form the metal layer **116m**. The metal layer **116m** may be formed concurrently with the forming of a metal layer (e.g., metal trace) of the metal interconnect structures **116b**, and using the same metal material as the metal interconnect structures **116b**.

[0094] A second dielectric layer **116L-2** of the package substrate lower dielectric layer **116** may then be formed on the first dielectric layer **116L-1** by any suitable deposition method (e.g., CVD, PVD, etc.). The opening **116c-02** may then be formed in the second dielectric layer **116L-2**, for example, by etching (e.g., wet etching, dry etching, etc.). The metal layer **116m** may serve as an etch stop layer in the etching of the opening **116c-02**.

[0095] FIG. 4B illustrates a vertical cross-sectional view of an intermediate structure in which dummy via layer **116c2** has been formed according to one or more embodiments. As illustrated in FIG. 4B, the dummy via layer **116c2** may be formed in the opening **116c-02** and on the metal layer **116m**. The dummy via layer **116c2** may be formed concurrently with the forming of a metal layer (e.g., metal via) of the metal interconnect structures **116b**, and using the same metal material as the metal interconnect structures **116b**.

[0096] FIG. 4C illustrates a vertical cross-sectional view of an intermediate structure in which an opening **116c-03** has been formed according to one or more embodiments. As illustrated in FIG. 4C, a third dielectric layer **116L-3** of the package substrate lower dielectric layer **116** may be formed on the second dielectric layer **116L-2** by any suitable deposition method (e.g., CVD, PVD, etc.). The opening **116c-03** may then be formed in the third dielectric layer **116L-3**, for example, by etching (e.g., wet etching, dry etching, etc.).

[0097] FIG. 4D illustrates a vertical cross-sectional view of an intermediate structure in which dummy via layer **116c3** has been formed according to one or more embodiments. As illustrated in FIG. 4D, the dummy via layer **116c3** may be formed in the opening **116c-03** and on the dummy via layer **116c2**. The dummy via layer **116c3** may be formed concurrently with the forming of a metal layer (e.g., metal trace) of the metal interconnect structures **116b**, and using the same metal material as the metal interconnect structures **116b**.

[0098] FIG. 5 illustrates a vertical cross-sectional view of a second alternative design of the semiconductor package **100**, according to one or more embodiments. As with the first alternative design in FIG. 3, the second alternative design may include stacked dummy vias **116d**. However, in the second alternative design, each stacked dummy via **116d** may include a dummy via layer **116c1**, dummy via layer **116c2** stacked on the dummy via layer **116c1**, and dummy via layer **116c3** stacked

on the dummy via layer **116c2**.

[0099] As illustrated in FIG. 5, the stacked dummy vias **116d** in FIG. 5 may have a thickness that is substantially the same as a thickness of the package substrate lower dielectric layer **116**. The dummy via layer **116cl** may be formed on the core **112** or on a metal layer on a surface of the core **112**. Further, in the alternative design (as in all of the embodiments of the semiconductor package **100**), the core **112** may be formed so as to accommodate the stacked dummy vias **116d**. In particular, the through vias **112a** may be formed outside the region for forming the stacked dummy vias **116d** so as not to interfere with the formation of the stacked dummy vias **116d**.

[0100] The stacked dummy vias **116d** in the second alternative design may be formed by a method that is substantially the same as the method described above with respect to FIGS. 4A-4D. That is, each of the dummy via layer **116c1**, dummy via layer **116c2** and dummy via layer **116c3** may be formed in different dielectric layers of package substrate lower dielectric layer **116**. In particular, the dummy via layer **116c1**, dummy via layer **116c2** and dummy via layer **116c3** may be formed concurrently with the forming of a metal layer (e.g., metal trace) of the metal interconnect structures **116b**, and using the same metal material as the metal interconnect structures **116b**.

[0101] FIG. 6A illustrates a vertical cross-sectional view of a third alternative design of the semiconductor package **100**, according to one or more embodiments. As illustrated in FIG. 6A, the third alternative design may include a dummy via array **116A** including a plurality of dummy vias **116c**. The dummy vias **116c** in the dummy via array **116A** may be arranged, for example, in columns and rows around the entire periphery of the interposer module. For example, in FIG. 6A, the dummy via array **116A** may include rows of dummy vias **116c** aligned in the x-direction, and three columns of the dummy vias **116c** in the y-direction (e.g., into the page). A centerline of the dummy via array **116A** may be substantially aligned with the inner edge **150a** of the stiffener ring. For example, in FIG. 6A, the centerline of the dummy via array **116A** may be coincident with the middle column of dummy vias **116c** in the dummy via array **116A**.

[0102] FIG. 6B is a plan view of the board-side surface of package substrate lower dielectric layer **116** for the third alternative design of the semiconductor package **100**, according to one or more embodiments. A location of the stiffener ring **150** and the interposer module **120** on the opposing surface (e.g., chip-side surface) of the package substrate **110** is indicated by shading. The outer periphery of the interposer module **120** and the inner edge **150a** and outer edge **150b** of the stiffener ring **150** are also indicated by a dashed line in FIG. 6B.

[0103] As illustrated in FIG. 6B, a layout of the dummy via array **116A** may be substantially similar to the layout of the dummy vias **116c** in FIG. 1B. The dummy via array **116A** may be located adjacent to solder balls **110c** of the BGA. In particular, the dummy via array **116A** may be formed between the inner set of solder balls **110c** under the interposer module **120**, and the outer set of solder balls **110c** under the stiffener ring **150**.

[0104] The dummy via array **116A** may be formed around an entire periphery of the interposer module **120** in the plan view. The inner edge **150a** may have a length L in the x-direction and a width W in the y-direction. Therefore, a total length of the inner edge **150a** may be given by $2(L+W)$.

[0105] In particular, the dummy via array **116A** may be formed under the inner edge **150a** of the stiffener ring **150** over the entire length of the inner edge **150a**. The dummy via array **116A** may be substantially aligned under the inner edge **150a** of the stiffener ring **150**. Thus, in a plan view the dummy via array **116A** may be arranged so as to have a ring shape substantially the same as a ring shape of the stiffener ring **150**.

[0106] As illustrated in FIG. 6B, the dummy vias **116c** in the dummy via array **116A** may extend away from the centerline by a distance D . Thus, a width (e.g., diameter) of the dummy via array **116A** in the x-direction and y-direction may be $2D$. In at least one embodiment, the distance D from the centerline of the dummy via array **116A** to an edge of the dummy via array **116A** may be

greater than 0.1 mm and less than 2 mm.

[0107] The dummy via array **116A** may be located in a region R having a width that is substantially the same as the width (2D) of the dummy via array **116A**. As with the dummy vias **116c** in FIG. **1B**, the area ratio R_a for the dummy vias **116c** in the dummy via array **116A** may be, for example, greater than 0.1 and less than 0.9.

[0108] FIG. **7** is a plan view of the board-side surface of package substrate lower dielectric layer **116** for a fourth alternative design of the semiconductor package **100**, according to one or more embodiments. The fourth alternative design may be substantially similar to the third alternative design in FIGS. **6A-6B**. However, in the fourth alternative design, the dummy via array **116A** is not necessarily formed continuously around the inner edge **150a** of the stiffener ring **150**. Instead, there may be portions of the region R around the inner edge **150a** where the dummy vias **116c** are not formed. It should be noted that the configuration of the dummy via array **116A** in FIG. **7** is not limiting, but there may be other configurations of the dummy via array **116A**. For example, there may be groups of dummy vias **116c** formed at one or more corners of the stiffener ring **150**, on one or more sides of the stiffener ring **150**, and so on. Further, the groups of dummy vias **116c** in the dummy via array **116A** may be formed in two columns, four columns, and so on.

[0109] FIGS. **8A-8C** illustrate alternative shapes (e.g., cross-sectional shapes) of the of the dummy vias **116c** in the dummy via array **116A**, according to one or more embodiments. It should be noted that the alternative shapes in FIGS. **8A-8C** may also be applied to the dummy vias **116c** in other embodiments (e.g., FIGS. **1A-1B**). In addition, the alternative shapes in FIGS. **8A-8C** are not intended to be limiting so that other shapes for the dummy vias **116c** may be within the scope of this disclosure. In addition, the dummy vias **116c** are not required to have one uniform shape, but may have two more different shapes in the semiconductor package **100**.

[0110] In particular, FIG. **8A** illustrates the dummy vias **116c** having a line shaped cross-section (e.g., looking down onto the board-side surface of the package substrate lower dielectric layer **116**), according to one or more embodiments. The line shape may be arranged so as to have a longitudinal direction in the x-direction and/or y-direction. FIG. **8B** illustrates the dummy vias **116c** having a rectangular shaped cross-section (e.g., square shaped), according to one or more embodiments. FIG. **8C** illustrates the dummy vias **116c** having a hexagonal shaped cross-section, according to one or more embodiments.

[0111] FIG. **9** is a flow chart illustrating a method of making a semiconductor package according to one or more embodiments. Step **910** includes forming a package substrate including a dummy via on a first side of the package substrate. Step **920** includes mounting an interposer module on a second side of the package substrate opposite the first side of the package substrate. Step **930** includes mounting a stiffener ring on the second side of the package substrate, such that an edge of the stiffener ring is substantially aligned with the dummy via.

[0112] Referring to FIGS. **1A-9**, a semiconductor package **100** may include a package substrate **110** including a dummy via **116c** on a first side of the package substrate **110**, an interposer module on a second side of the package substrate **110** opposite the first side of the package substrate **110**, and a stiffener ring **150** on the second side of the package substrate **110** and including an edge that is substantially aligned with the dummy via **116c**. In one embodiment, the dummy via **116c** may include a plurality of dummy vias **116c** constituting a dummy via array **116A**. In one embodiment, the centerline of the dummy via array **116A** may be substantially aligned with the edge of the stiffener ring **150**. In one embodiment, the distance from the centerline of the dummy via array **116A** to an edge of the dummy via array **116A** may be greater than 0.1 mm and less than 2 mm. In one embodiment, the ratio R_a of an area of the dummy via array **116A** to an area of a region of the package substrate **110** including the dummy array via is greater than 0.1 and less than 0.9. In one embodiment, the plan view the dummy via array **116A** may include a ring shape substantially the same as a ring shape of the stiffener ring **150**. In one embodiment, the package substrate **110** may include a core **112**, a package substrate upper dielectric layer **114** on the core and on the first side of

the package substrate **110**, and a package substrate lower dielectric layer **116** on the core and on the second side of the package substrate **110**, wherein the dummy via **116c** is located in the package substrate lower dielectric layer **116**. In one embodiment, the height of the dummy via **116c** may be less than a thickness of the package substrate lower dielectric layer **116**. In one embodiment, the semiconductor package **100** may further include a ball grid array on the second side of the package substrate **110**, wherein the dummy via **116c** is located adjacent to the ball grid array. In one embodiment, the dummy via **116c** may include a plurality of stacked dummy via layers **116c1**, **116c2**, **116c3**. A shape of a cross-section of the dummy via **116c** may include a circular shape, a rectangular shape or a hexagonal shape. In one embodiment, the edge of the stiffener ring **150** may include an inner edge of the stiffener ring **150**, and the dummy via **116c** may be located between the interposer module and the stiffener ring **150** in a plan view.

[0113] Referring to FIGS. **1A-9**, a method of forming a semiconductor package **100** may include forming a package substrate **110** including a dummy via **116c** on a first side of the package substrate **110**, mounting an interposer module on a second side of the package substrate **110** opposite the first side of the package substrate **110**, and mounting a stiffener ring **150** on the second side of the package substrate **110**, such that an edge of the stiffener ring **150** is substantially aligned with the dummy via **116c**. In one embodiment, the forming of the dummy via **116c** may include forming a plurality of dummy vias **116c** constituting a dummy via array **116A**. In one embodiment, the forming of the plurality of dummy vias **116c** may include substantially aligning a centerline of the dummy via array **116A** with the edge of the stiffener ring **150**. In one embodiment, the substantially aligning of the centerline of the dummy via array **116A** may include locating the centerline of the dummy via array **116A** such that a distance from the centerline to an edge of the dummy via array **116A** is greater than 0.1 mm and less than 2 mm. In one embodiment, the forming of the plurality of dummy vias **116c** may include forming the plurality of dummy vias **116c** such that a ratio R_a of an area of the dummy via array **116A** to an area of a region R of the package substrate **110** may include the dummy array via is greater than 0.1 and less than 0.9. In one embodiment, the forming of the package substrate **110** may include forming a core, forming a package substrate upper dielectric layer **114** on the core **112** and on the first side of the package substrate **110**, forming a package substrate lower dielectric layer on the core **112** and on the second side of the package substrate **110**, and forming the dummy via **116c** in the package substrate lower dielectric layer. In one embodiment, the method may further include forming a ball grid array on the first side of the package substrate lower dielectric layer such that the dummy via **116c** is located adjacent to the ball grid array.

[0114] Referring to FIGS. **1A-9**, a semiconductor package **100** may include a substrate **110** including a core **112** including a plurality of vias **112a**, an upper dielectric layer **114** on the core **112** and on the first side of the substrate **110**, and may include a plurality of upper metal interconnect structures **114b**, a lower dielectric layer **116** on the core **112** and on the second side of the substrate **110**, and may include a plurality of lower metal interconnect structures **116b** electrically coupled to the plurality of upper metal interconnects **114b** by the plurality of vias **112a**, and a dummy via array **116A** located in the lower dielectric layer **116** adjacent to the plurality of lower metal interconnect structures **116b**. The semiconductor package **100** may further include an interposer module **120** on a second side of the substrate **110** opposite the first side of the substrate **110**, a stiffener ring **150** on the second side of the substrate **110** and including an edge that is substantially aligned with a centerline of the dummy via array **116A**, wherein a distance from the centerline of the dummy via array **116A** to an edge of the dummy via array **116A** is greater than 0.1 mm and less than 2 mm, and wherein a height of the dummy via array **116A** is less than a thickness of the lower dielectric layer **116**, and a ball grid array on a bottom surface of the lower dielectric layer **116** of the substrate **110**, wherein the dummy via array **116A** is located adjacent to the ball grid array in the bottom surface of the lower dielectric layer **116** of the substrate **110**.

[0115] The foregoing outlines features of several embodiments so that those skilled in the art may

better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A package substrate, comprising: a core; a lower dielectric layer on a first side of the core and including a dummy via; and an upper dielectric layer on a second side of the core and including a semiconductor module mounting region and stiffener ring mounting region.
2. The package substrate of claim 1, wherein the dummy via is substantially aligned with an edge of the stiffener ring mounting region.
3. The package substrate of claim 1, wherein a center of the dummy via is substantially aligned with an edge of the stiffener ring mounting region.
4. The package substrate of claim 1, wherein the dummy via comprises a plurality of dummy vias constituting a dummy via array.
5. The package substrate of claim 4, wherein a centerline of the dummy via array is substantially aligned with the edge of the stiffener ring mounting region.
6. The package substrate of claim 4, wherein the stiffener ring mounting region comprises a ring shape and the dummy via array comprises a ring shape substantially the same as the ring shape of the stiffener ring mounting region.
7. The package substrate of claim 1, wherein a height of the dummy via is less than a thickness of the lower dielectric layer.
8. The package substrate of claim 1, wherein the dummy via comprises a plurality of stacked dummy via layers.
9. The package substrate of claim 1, wherein the lower dielectric layer further comprises a ball grid array mounting region on the second side of the package substrate, and the dummy via is located adjacent the ball grid array mounting region.
10. A package structure, comprising: a package substrate comprising a dummy via on a first side of the package substrate; a semiconductor module on a second side of the package substrate; and a stiffener ring adjacent the semiconductor module on the second side of the package substrate.
11. The package structure of claim 10, wherein the dummy via is substantially aligned with an edge of the stiffener ring.
12. The package structure of claim 10, wherein a center of the dummy via is substantially aligned with an edge of the stiffener ring.
13. The package structure of claim 10, wherein the dummy via comprises a plurality of dummy vias constituting a dummy via array.
14. The package structure of claim 13, wherein a centerline of the dummy via array is substantially aligned with the edge of the stiffener ring.
15. The package structure of claim 14, wherein a distance from the centerline of the dummy via array to an edge of the dummy via array is greater than 0.1 mm and less than 2 mm.
16. The package structure of claim 13, wherein a ratio of an area of the dummy via array to an area of a region of the package substrate including the dummy via array is greater than 0.1 and less than 0.9.
17. The package structure of claim 10, wherein the package substrate comprises a core, a lower dielectric layer on the core and a second dielectric layer on a side of the core opposite the lower dielectric layer, the dummy via is in the lower dielectric layer and a height of the dummy via is less

than a thickness of the lower dielectric layer.

18. The package structure of claim 10, wherein the dummy via comprises a plurality of stacked dummy via layers.

19. The package structure of claim 10, further comprising: a ball grid array on the first side of the package substrate, wherein the dummy via is located adjacent the ball grid array.

20. A method of making a package structure, the method comprising: forming a package substrate comprising a dummy via on a first side of the package substrate; attaching a semiconductor module to a second side of the package substrate; and attaching a stiffener ring to the second side of the package substrate.
